

URUMQI DIWOPU INTL, China

WMO: 514635

Lat: 43.907N

Lon: 87.474E

Elev: 2125

StdP: 13.60

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.4	(c) -7.3	(d) -16.2	(e) 2.5	(f) -8.6	(g) -11.3	(h) 3.2	(i) -5.9	(j) 9.9	(k) 18.8	(l) 7.8	(m) 14.6	(n) 1.9	(o) 320	(p) 0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	96.7	64.9	93.5	64.1	91.3	63.4	68.4	86.5	67.0	85.2	65.7	84.1	7.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	96.5	72.0	61.1	87.1	72.4	59.2	81.1	73.9	34.0	86.5	32.7	84.4	31.7	83.3	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.4	(b) 12.8	(c) 10.7	(e) -14.3	(f) 103.1	(g) 5.4	(h) 2.8	(i) -18.2	(j) 105.1	(k) -21.3	(l) 106.7	(m) -24.4	(n) 108.3	(o) -28.4	(p) 110.4		
			DB													
			WB													
				(e) -12.6	(f) 71.0	(g) 8.2	(h) 2.9	(i) -18.5	(j) 73.1	(k) -23.3	(l) 74.8	(m) -27.9	(n) 76.5	(o) -33.8	(p) 78.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	6.4	14.0	34.8	56.0	66.9	76.3	79.6	77.2	67.0	51.1	32.4	13.6		
	DBStd	27.12	8.39	8.37	12.33	9.77	8.26	6.52	5.96	6.61	8.33	9.09	10.19	8.75		
	HDD50	4669	1351	1007	491	54	2	0	0	0	3	100	532	1129		
	HDD65	7643	1816	1427	937	292	78	5	1	4	73	437	979	1594		
	CDD50	3977	0	0	20	234	525	790	916	844	512	134	2	0		
	CDD65	1474	0	0	0	22	136	345	452	382	132	5	0	0		
	CDH74	15723	0	0	2	183	1274	3757	5229	4115	1129	34	0	0		
	CDH80	6829	0	0	0	38	423	1642	2492	1860	370	4	0	0		
(14) Wind	WSAvg	4.7	2.5	2.8	4.0	6.0	6.4	6.2	6.1	5.9	5.6	4.6	3.5	2.8		
(15) Precipitation	PrecAvg	9.1	0.4	0.4	0.5	1.0	1.1	1.1	1.5	1.0	0.6	0.6	0.6	0.5		
	PrecMax	12.2	0.7	1.1	1.2	2.4	3.1	2.4	3.5	2.9	1.7	1.5	1.1	1.3		
	PrecMin	5.2	0.1	0.0	0.1	0.0	0.1	0.3	0.2	0.1	0.1	0.0	0.1	0.2		
	PrecStd	2.0	0.2	0.2	0.3	0.6	0.7	0.6	0.8	0.7	0.4	0.4	0.3	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	36.0	69.7	85.6	92.9	97.1	102.1	100.2	93.3	78.5	59.3	35.7		
		MCWB	29.7	31.9	50.2	55.5	58.8	63.9	65.5	64.6	61.8	54.5	47.3	32.4		
	2%	DB	26.2	32.0	62.9	79.0	88.0	94.2	97.0	95.3	88.1	73.0	53.7	31.9		
		MCWB	24.4	29.0	47.4	53.0	58.1	63.4	65.6	64.5	60.1	53.1	43.6	30.0		
	5%	DB	21.4	28.6	57.6	75.0	84.5	91.5	93.6	93.0	84.2	68.4	49.7	28.5		
		MCWB	20.2	26.5	45.7	52.3	57.3	62.8	65.1	64.3	58.4	51.1	41.7	27.1		
	10%	DB	18.0	25.2	53.2	71.4	80.9	88.2	91.3	89.5	80.5	64.7	46.0	26.3		
		MCWB	16.9	23.6	43.8	50.6	56.4	62.3	64.5	63.5	57.3	49.5	39.8	25.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.1	33.4	51.3	58.0	62.9	69.2	72.2	70.0	65.5	56.4	48.2	32.8		
		MCDB	31.4	35.1	65.6	77.2	81.9	89.7	90.7	86.7	85.6	73.3	59.4	34.8		
	2%	WB	24.1	29.5	48.4	55.6	61.2	66.7	68.9	67.8	62.0	54.0	44.5	29.9		
		MCDB	25.5	31.6	61.1	72.4	80.4	85.3	86.6	86.3	80.9	69.0	51.4	31.2		
	5%	WB	20.3	26.5	46.4	53.6	59.8	65.3	67.4	66.3	60.2	52.3	42.0	26.8		
		MCDB	21.4	28.4	57.0	70.1	78.6	84.2	85.4	84.5	79.8	66.4	48.3	28.1		
	10%	WB	17.1	24.0	43.9	52.1	58.3	63.9	66.3	65.0	58.6	50.3	39.7	24.5		
		MCDB	18.1	25.5	52.1	67.1	76.4	82.8	84.4	83.3	77.4	63.3	45.1	25.7		
(35) Mean Daily Temperature Range	MDBR		13.3	13.8	14.2	18.6	19.0	18.3	17.8	18.3	18.6	16.7	12.1	12.1		
	5% DB	MCDBR	16.8	16.5	20.3	22.2	21.8	21.1	20.3	20.9	21.1	19.8	17.5	14.4		
		MCWBR	14.8	13.5	10.2	9.2	8.3	7.8	7.8	8.0	8.6	9.4	10.5	12.0		
	5% WB	MCDBR	16.3	15.4	19.1	19.7	19.2	18.0	17.7	18.3	19.1	18.7	16.3	13.8		
		MCWBR	14.5	12.8	10.2	8.8	8.3	7.3	7.4	7.6	8.3	9.5	10.3	11.7		
(40) Clear-Sky Solar Irradiance	taub		0.348	0.401	0.458	0.513	0.500	0.510	0.490	0.470	0.445	0.430	0.386	0.350		
	taud		2.064	1.973	1.903	1.826	1.900	1.937	2.030	2.066	2.055	2.020	2.077	2.106		
	Ebn at Noon		232	239	242	239	248	246	249	249	244	226	215	218		
	Edh at Noon		34	44	54	63	60	58	53	49	46	42	33	30		
(44) All-Sky Solar Radiation	RadAvg		620	911	1328	1753	2045	2090	2042	1869	1593	1115	683	499		
	RadStd		28	53	85	109	102	100	93	81	52	62	56	45		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	-1.24	N/A	N/A	+236	+162

Nomenclature: See separate page